

Runbo Li

✉ runbo.li26@imperial.ac.uk

✉ runbo.li.carey@gmail.com

🌐 <https://runbolicarey.com>

Education

- 2026 – **B.Sc.** Department of Mathematics, Imperial College London, South Kensington Campus, London SW7 2AZ, UK
- 2023 – 2026 International Curriculum Center (ICC), The High School Affiliated to Renmin University of China, Beijing, China

Publications

- 5 **Runbo Li.** A note on variants of Buchstab's identity. *Bulletin of the Belgian Mathematical Society – Simon Stevin.* to appear.
- 4 **Runbo Li.** On the largest prime factor of integers in short intervals. *Cambridge Journal for Junior Scientists.* 3(1): 1–16, 2026. <https://doi.org/10.4310/CJJS.260205233711>.
- 3 **Runbo Li.** A remark on the distribution of $\sqrt{p}$ modulo one involving primes of special type. *Hiroshima Mathematical Journal.* 56(1): 57–60, 2026. <https://doi.org/10.32917/h2024011>.
- 2 **Runbo Li.** On Chen's theorem, Goldbach's conjecture and almost prime twins II. *Mathematical Reports.* 28(1–2): 39–61, 2026. <https://doi.org/10.59277/mrar.2026.28.78.1.2.39>.
- 1 **Runbo Li.** A remark on the distribution of $\sqrt{p}$ modulo one involving primes of special type II. *European Journal of Mathematics.* 11:85, 2025. <https://doi.org/10.1007/s40879-025-00877-y>.

Preprints

- 35 **Runbo Li.** On the largest prime factor of integers in short intervals IV. *Cambridge Open Engage.* 10.33774/coe-2026-24kjf.
- 34 **Runbo Li.** The number of Piatetski-Shapiro primes in short intervals. *Cambridge Open Engage.* 10.33774/coe-2026-qpr94.
- 33 **Runbo Li.** Primes in arithmetic progressions to large moduli and refinements of Harman's sieve. *arXiv.* arXiv:2602.20917v6.

- 32 **Runbo Li.** On the largest prime factor of integers in short intervals III. *arXiv.* [*arXiv:2601.00910v1.*](#)
- 31 **Runbo Li.** Sums of two squares of primes in short intervals. *Cambridge Open Engage.* [*10.33774/coe-2025-15m3h.*](#)
- 30 **Runbo Li.** Bombieri–Vinogradov Theorem in shorter intervals. *Preprints.* [*preprints202511.0089.v1.*](#)
- 29 **Runbo Li.** Primes in almost all short intervals II. *Cambridge Open Engage.* [*10.33774/coe-2025-jrnjl.*](#)
- 28 **Runbo Li.** On Chen's theorem, Goldbach's conjecture and almost prime twins III. *Cambridge Open Engage.* [*10.33774/coe-2025-5kl50.*](#)
- 27 **Runbo Li.** An average Brun–Titchmarsh theorem and shifted primes with a large prime factor. *arXiv.* [*arXiv:2508.18285v3.*](#)
- 26 **Runbo Li.** On the weighted AM-GM inequality and refined inequalities between arithmetic functions. *Preprints.* [*preprints202511.0246.v1.*](#)
- 25 **Runbo Li.** On the exceptional set in the abc conjecture. *arXiv.* [*arXiv:2507.02885v1.*](#)
- 24 **Runbo Li.** On the largest prime factor of integers in short intervals II. *Cambridge Open Engage.* [*10.33774/coe-2025-xnbjq-v2.*](#)
- 23 **Runbo Li.** On Chen's theorem, Goldbach's conjecture and almost prime twins II. *arXiv.* [*arXiv:2405.05727v4.*](#)
- 22 **Runbo Li.** On the largest prime factor of quadratic polynomials. *arXiv.* [*arXiv:2406.07575v2.*](#)
- 21 **Runbo Li.** A remark on the distribution of $\sqrt{p}$ modulo one involving primes of special type II. *arXiv.* [*arXiv:2401.01351v3.*](#)
- 20 **Runbo Li.** Largest square divisors of shifted primes. *arXiv.* [*arXiv:2505.23779v2.*](#)
- 19 **Runbo Li.** On the generalized Dirichlet divisor problem. *Preprints.* [*preprints202505.1432.v3.*](#)
- 18 **Runbo Li.** On almost primes in Piatetski-Shapiro sequences. *arXiv.* [*arXiv:2505.09634v1.*](#)
- 17 **Runbo Li.** Primes in arithmetic progressions to smooth moduli: A minorant version. *arXiv.* [*arXiv:2505.09629v3.*](#)

- 16 **Runbo Li.** On prime-producing sieves and distribution of $\alpha p - \beta \bmod 1$. *arXiv.* [*arXiv:2504.13195v3*](#).
- 15 **Runbo Li.** A note on variants of Buchstab's identity. *arXiv.* [*arXiv:2504.07974v1*](#).
- 14 **Runbo Li.** The number of primes in short intervals and numerical calculations for Harman's sieve. *arXiv.* [*arXiv:2308.04458v8*](#).
- 13 **Runbo Li.** On the Piatetski-Shapiro prime number theorem II. *Preprints.* [*preprints202504.1165.v2*](#).
- 12 **Runbo Li.** On the Piatetski-Shapiro prime number theorem. *Preprints.* [*preprints202504.1190.v1*](#).
- 11 **Runbo Li.** On the largest prime factor of integers in short intervals. *Preprints.* [*preprints202504.1212.v2*](#).
- 10 **Runbo Li.** Primes in almost all short intervals. *arXiv.* [*arXiv:2407.05651v6*](#).
- 9 **Runbo Li.** On the primitive divisors of quadratic polynomials. *arXiv.* [*arXiv:2406.07575v1*](#).
- 8 **Runbo Li.** On Chen's theorem, Goldbach's conjecture and almost prime twins. *arXiv.* [*arXiv:2405.05727v3*](#).
- 7 **Runbo Li.** On a conjecture involving twin practical numbers. *Preprints.* [*preprints202504.1211.v2*](#).
- 6 **Runbo Li.** A remark on large even integer of the form $p + P_3$. *arXiv.* [*arXiv:2403.09691v1*](#).
- 5 **Runbo Li.** Hybrid estimation of exponential sums, exceptional characters and primes in short intervals. *arXiv.* [*arXiv:2401.11139v3*](#).
- 4 **Runbo Li.** Remarks on additive representations of natural numbers. *arXiv.* [*arXiv:2309.03218v7*](#).
- 3 **Runbo Li.** A remark on the distribution of $\sqrt{p}$ modulo one involving primes of special type. *arXiv.* [*arXiv:2401.01351v1*](#).
- 2 **Runbo Li.** On the upper and lower bound orders of almost prime triples. *arXiv.* [*arXiv:2401.01348v1*](#).
- 1 **Runbo Li.** On some problems of primes with the floor function. *arXiv.* [*arXiv:2308.16301v1*](#).

Manuscripts

- 3 **Runbo Li.** Waring–Goldbach Problem: One square, four fifth powers and twenty-two seventh powers.
- 2 **Runbo Li.** Waring–Goldbach Problem: One square, four fifth powers and fifteen sixth powers.
- 1 **Runbo Li.** Primes in almost all short intervals III.

Presentations

- 6 Primes in arithmetic progressions to large moduli. *The 4th Workshop on Cute Math.* Online. 26 July 2026.
- 5 Primes in short intervals. *Nankai University Additive Number Theory Seminar.* Online. 4 December 2025.
- 4 On Bertrand's Postulate. *Tianjin Nankai High School: Student Research Experience Sharing and Shiing-Shen Chern Class Research Group Launch Meeting.* Online. 21 November 2025.
- 3 Primes in short intervals. *The 3rd Workshop on Cute Math.* Online. 21 July 2025.
- 2 On the Goldbach's conjecture. *2025 'Haidian π Mathematics Festival' Closing Ceremony Research Presentation.* The High School Affiliated to Beijing Institute of Technology. 4 July 2025.
- 1 On Chen's theorem, Goldbach's conjecture and applications of sieve methods. *S.-T. Yau High School Science Award (Mathematics) Global Final Research Report.* Tsinghua University. 7 December 2024.

Attended Seminars

- 2026 2026 Chen Jing-Run Prize Conference (Beijing, China)
- 2026 Forum for young scholar in number theory and related fields (Xiamen, China)
- 2024 2024 Number Theory Workshop at Peking (The 9th National Conference on Number Theory) (Beijing, China)
- 2024 International Number Theory Conference in Commemoration of Chengdong Pan (Jinan, China)